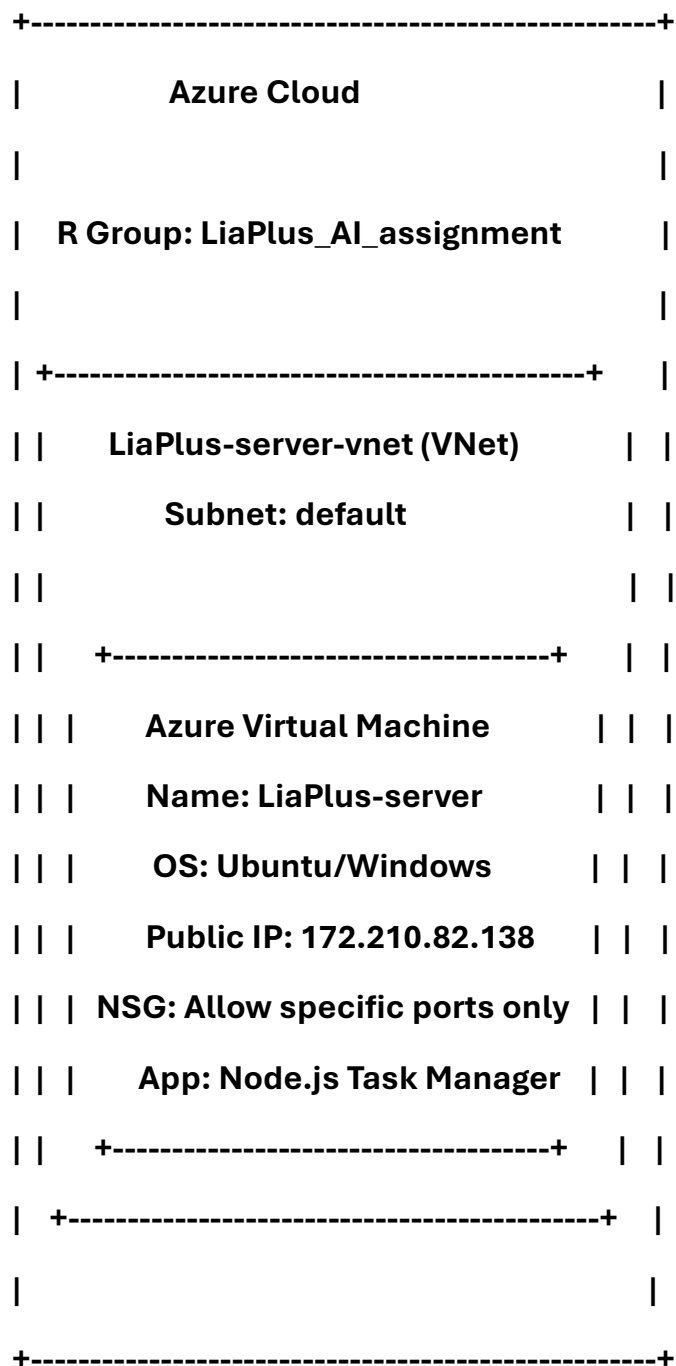


Task 1: Set up a simple web application (Node.js) and deploy it on an Azure Virtual Machine. Ensure that the application is accessible via a public URL.

Architecture Diagram:

Access the application on: <http://172.210.82.138:3000>



Steps to Deploy the Application:

1. Create Resource Group

Home > Resource groups > LiaPlus_AI_assignment

Resources Recommendations

Filter for any field... Type equals all Location equals all Add filter

Showing 1 to 7 of 7 records. Show hidden types No grouping List view

Name	Type	Location
LiaPlus-server	Virtual machine	East US
LiaPlus-server-ip	Public IP address	East US
LiaPlus-Server-Sg	Network security group	East US
LiaPlus-server-vnet	Virtual network	East US
liaplus-server795_z3	Network Interface	East US
LiaPlus-server_disk1_d3273bc8c6fd4510b8e4647d18ae81f2	Disk	East US

2. Create a Network Security Group (NSG) for the Server(VM)

Home > Resource groups > LiaPlus_AI_assignment > LiaPlus-Server-Sg

Network security group

Move Delete Refresh Give feedback

Essentials

Resource group (move): LiaPlus_AI_assignment Custom security rules: 3 inbound, 0 outbound

Location: East US Associated with: 0 subnets, 1 network interfaces

Subscription (move): Azure for Students

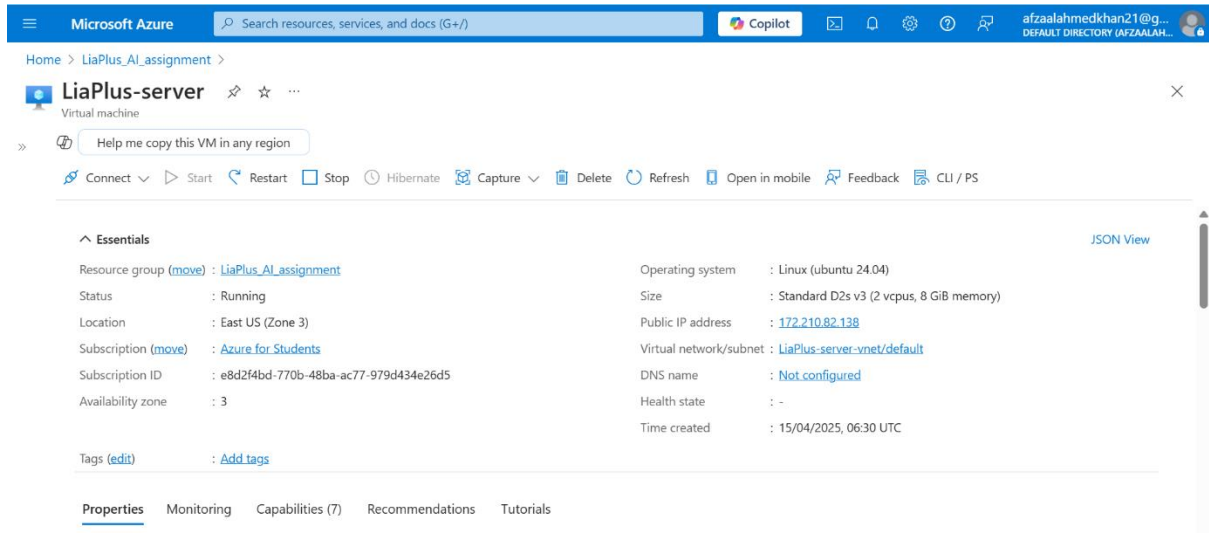
Subscription ID: e8d2f4bd-770b-48ba-ac77-979d434e26d5

Tags (edit): Add tags

Filter by name Port == all Protocol == all Source == all Destination == all Action == all

Priority	Name	Port	Protocol	Source	Destination	Action
100	AllowAnyCustom3000Inb...	3000	TCP	Any	Any	Allow
110	AllowAnyCustom80Inbou...	80	TCP	Any	Any	Allow
120	AllowMypAddressCusto...	22	TCP	14.97.132.203	172.210.82.138,10.0.0.4	Allow
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow

3. Create the Virtual Machine that hosts the application



4. Deploy the Application

1. SSH into the VM :

```
ssh -i Downloads/LiaPlus_ssh_key.pem afzaal@172.210.82.138
```

2. Install dependencies

```
sudo apt update && sudo apt upgrade -y
```

```
sudo apt install -y nodejs npm git
```

```
node -v && npm -v
```

3. Clone the app repo

```
git clone https://github.com/AFZAALAHMEDKHAN/Task_manager_API.git
```

```
cd Task_manager_API/
```

4. And finally, deploying the application

```
afzaal@LiaPlus-server: ~/Task_manager_API
afzaal@LiaPlus-server:~$ git clone https://github.com/AFZAALAHMEDKHAN/Task_manager_API.git
Cloning into 'Task_manager_API'...
remote: Enumerating objects: 28, done.
remote: Counting objects: 100% (28/28), done.
remote: Compressing objects: 100% (21/21), done.
remote: Total 28 (delta 1), reused 28 (delta 1), pack-reused 0 (from 0)
Receiving objects: 100% (28/28), 33.23 KiB | 11.08 MiB/s, done.
Resolving deltas: 100% (1/1), done.
afzaal@LiaPlus-server:~$ cd Task_manager_API/
afzaal@LiaPlus-server:~/Task_manager_API$ ls
app.js controllers db errors middleware models package-lock.json package.json public routes
afzaal@LiaPlus-server:~/Task_manager_API$ vim .env
afzaal@LiaPlus-server:~/Task_manager_API$
afzaal@LiaPlus-server:~/Task_manager_API$
afzaal@LiaPlus-server:~/Task_manager_API$
afzaal@LiaPlus-server:~/Task_manager_API$ npm install
added 208 packages, and audited 209 packages in 4s
25 packages are looking for funding
  run `npm fund` for details
3 high severity vulnerabilities
To address all issues, run:
  npm audit fix
Run `npm audit` for details.
afzaal@LiaPlus-server:~/Task_manager_API$
afzaal@LiaPlus-server:~/Task_manager_API$
afzaal@LiaPlus-server:~/Task_manager_API$ npm start
> jobs@1.0.0 start
> node app.js
Attempting to connect to database...
Connected to the DB...
Database connection successful.
Server listening on port 3000
^C
afzaal@LiaPlus-server:~/Task_manager_API$ |
```

5. While testing, to keep the application running even after closing the terminal, using the command : nohup npm start &

```
Last login: Tue Apr 15 06:50:17 2025 from 14.97.132.203
afzaal@LiaPlus-server:~$ cd Task_manager_API/
afzaal@LiaPlus-server:~/Task_manager_API$ nohup npm start &
[1] 8563
afzaal@LiaPlus-server:~/Task_manager_API$ nohup: ignoring input and appending ou
tput to 'nohup.out'
afzaal@LiaPlus-server:~/Task_manager_API$ exit
logout
Connection to 172.210.82.138 closed.
```

6. The application is managed using PM2 in the CI/CD GitHub workflow to ensure it runs continuously and restarts automatically on failure or VM reboot.

